

ON A PROBLEM OF REGRESSION

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1. Introduction and Summary

In this article we will consider some necessary and sufficient conditions for the existence of rational regression of one stochastic matrix on a number of stochastic matrices. We will also discuss some characterization theorems for the multivariate normal distribution by using the linear regression properties.

Characterization of probability distributions by using regression properties has been investigated by many authors from time to time. Lukacs (1962) studied the characterization of the Poisson distribution by using the property of constant regression of $K_{p+r} - K_p$ on K_1 where K_p is the K -statistic of order p . Characterization of a univariate normal distribution by using linearity of regression and homoscedasticity, is discussed in Lukacs & Laha (1964). Regression problems in connection with symmetric stable distributions with the same exponent, was investigated by Neyman (1951) and Wolfowitz (1953). A detailed reference on the study of regression problems in connection with identifiability of distributions, structural set up of stochastic variables, symmetric stable laws, nature of the distribution and characteristic function and characterization of distributions, in univariate cases, is given in Lukacs & Laha (1964).

In what follows, all the stochastic variables considered are assumed to be non-degenerate in the sense that all the probability mass is not concentrated at one point. A matrix will be called finite if any norm is finite.

2. Rational Regression

Let $X, X_1, \dots, X_n$ be $n+1$ stochastic square matrices of the same

finite order. Let $E(X X_1^{r_1} \dots X_n^{r_n})$ for $r_1 + \dots + r_n \leq p$ and

$E(X_1^{s_1} \dots X_n^{s_n})$ for $s_1 + \dots + s_n \leq q$ exist, where E denotes 'mathematical expectation' and r 's and s 's are non-negative integers. Here a matrix is said to exist in the sense that any norm is finite and a sufficient condition is that every element exists. Let $P(X_1, \dots, X_n)$ and $Q(X_1, \dots, X_n)$ be polynomials in the matrices $X_1, \dots, X_n$ and of degrees p and q respectively.

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Theorem 1. A necessary and sufficient condition that the regression of X on $X_1, \dots, X_n$, that is, $E(X|X_1, \dots, X_n)$ is of the form

$$E(X|X_1, \dots, X_n)P(X_1, \dots, X_n) = Q(X_1, \dots, X_n) \text{ for all given } X_1, \dots, X_n,$$

is that,

$$E[X P(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}] = E[Q(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}]$$
for all real T_j , where T_j are square matrices of arbitrary real constants and $\operatorname{tr}(\cdot) = \text{trace } (\cdot) = \text{sum of the diagonal elements in } (\cdot)$, $i = (-1)^{1/2}$ and $E(Z) = (EZ_{rs})$ where $Z = (Z_{rs})$ is any matrix.

Proof

Let $E(X|X_1, \dots, X_n)P(X_1, \dots, X_n) = Q(X_1, \dots, X_n)$ for all given X_j 's. Multiply both sides by $e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}$ and then take expectations. Then $E[XP(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}]$

$$= E[Q(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}]$$

since $E(U) = E[E(U|V)]$ for any sets of stochastic variables U and V . In order to prove sufficiency we assume that,

$$E[X P(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}] = E[Q(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}].$$

That is, $E[E(X P e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j} | X_1, \dots, X_n)]$

$$= E[E(Q e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j} | X_1, \dots, X_n)].$$

Hence, $E e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j} \{E[(X P - Q) | X_1, \dots, X_n]\} = 0$.

(In what follows '0' denotes a null matrix).

Since all the elements in the matrix $E[(X P - Q) | X_1, \dots, X_n]$ are assumed to exist, by taking Fourier transform applied to functions of bounded variation we get, $E[(X P - Q) | X_1, \dots, X_n] = 0$

Therefore,

$$E(X|X_1, \dots, X_n) P = Q \text{ for all given } X_1, \dots, X_n.$$

This completes the proof.

Theorem 2

Let $X, X_1, \dots, X_n$ be stochastic row vectors of order k each with finite expected values. The necessary and sufficient condition that, $E(X|X_1, \dots, X_n) = \sum_{j=1}^n \alpha_j X_j$ for all given $X_1, \dots, X_n$, is that,

$$E(X e^{i \sum_{j=1}^n t_j X_j'}) = \sum_{j=1}^n \alpha_j E(X_j e^{i \sum_{j=1}^n t_j X_j'})$$
where t_j are row vectors of real constants and X_j' is the transpose of X_j and α_j are constants.

The proof follows in exactly similar line as in theorem 1.

Theorem 3

Let $X, X_1, \dots, X_n$ be $n+1$ symmetric stochastic matrices of the same finite order. Let $M(T_1, T_2, \dots, T_n) = E e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)}$ where T 's are square matrices of real constants. Let $P(X_1, \dots, X_n)$ and $Q(X_1, \dots, X_n)$ be polynomials in $X_1, \dots, X_n$ of degrees p and q respectively. Further let $P(iX_1, \dots, iX_n) = i^{r-1} P(X_1, \dots, X_n)$ and $Q(iX_1, \dots, iX_n) = i^r Q(X_1, \dots, X_n)$ for some $r \geq 1$. Let $E(X X_1^{r_1} \dots X_n^{r_n})$ for $r_1 + \dots + r_n \leq p$ and $E(X_1^{s_1} \dots X_n^{s_n})$ for $s_1 + \dots + s_n \leq q$ exist. Then the necessary and sufficient condition that the regression of X on $X_1, \dots, X_n$, is of the form,

$E(X | X_1, \dots, X_n) P(X_1, \dots, X_n) = Q(X_1, \dots, X_n)$ for all given $X_1, \dots, X_n$ is that,

$$\left[\frac{\partial}{\partial T} P\left(\frac{\partial}{\partial T_1}, \dots, \frac{\partial}{\partial T_n}\right) M(T_1, T_2, \dots, T_n) \right]_{T=0} = Q\left(\frac{\partial}{\partial T_1}, \dots, \frac{\partial}{\partial T_n}\right) M(0, T_1, \dots, T_n)$$

where for example the operator $\frac{\partial}{\partial T} M = \left(\frac{\partial M}{\partial T_{uv}} \right)$ and

$$\left(\frac{\partial}{\partial T_j} \right)^r = \frac{\partial^r}{\partial T_j^r} \text{ for } j = 1, 2, \dots, n.$$

Proof

$$\begin{aligned} \left[\frac{\partial}{\partial T} M(T_1, T_2, \dots, T_n) \right]_{T=0} &= E \left(\frac{\partial}{\partial T} e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)} \Big|_{T=0} \right) \\ &= i E(X e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}). \end{aligned}$$

(Since X is symmetric $\frac{\partial}{\partial T} e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)}$

$$\begin{aligned} &= i X^T e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)} = i X e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)}, \\ P\left(\frac{\partial}{\partial T_1}, \frac{\partial}{\partial T_2}, \dots, \frac{\partial}{\partial T_n}\right) M(0, T_1, \dots, T_n) &= i^{r-1} E[P(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}] \end{aligned}$$

for some $r \geq 1$. Therefore,

$$\left[\frac{\partial}{\partial T} P\left(\frac{\partial}{\partial T_1}, \dots, \frac{\partial}{\partial T_n}\right) M(T_1, T_2, \dots, T_n) \right]_{T=0} = i^r E[X P(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}]$$

and

$$Q\left(\frac{\partial}{\partial T_1}, \dots, \frac{\partial}{\partial T_n}\right) M(0, T_1, \dots, T_n) = i^r E[Q(X_1, \dots, X_n) e^{i \operatorname{tr} \sum_{j=1}^n T_j X_j}].$$

Hence the proof follows by theorem 1.

If $E e^{i \operatorname{tr}(TX + \sum_{j=1}^n T_j X_j)}$ exists for the stochastic matrices $X, X_1, \dots, X_n$ then the condition on the indices of the various

terms in the polynomials, that is, $P(iX_1, \dots, iX_n) = i^{r-1} P(X_1, \dots, X_n)$, and $Q(iX_1, \dots, iX_n) = i^r Q(X_1, \dots, X_n)$, can be dropped.

3. Some characterization theorems

In this section we will consider the problem of characterizing the multivariate normal distribution by using the regression properties.

Theorem 4

Let $X_1, \dots, X_n$ be n independently but not necessarily identically distributed stochastic column vectors of order k each with finite covariance matrices. Let $U = a_1 X_1 + \dots + a_n X_n$ and $V = b_1 X_1 + \dots + b_n X_n$ where a 's and b 's are scalars and $a_j b_j \neq 0$ for $j = 1, \dots, n$. If $E(V|U) = A + b U$ and $\text{Var}(V|U) = B$ is positive definite and independent of U , then X_j for which $a_j \neq b b_j$ has a multivariate normal distribution.

Proof

Let t be an arbitrary column vector of constants. Consider the stochastic variables $t'U$ and $t'V$. Since $X_1, \dots, X_n$ are stochastically independent, $t'X_1, \dots, t'X_n$ are also stochastically independent. Since $E(V|U) = A + b U$ and $\text{Var}(V|U) = B$ we have, $E(t'V|t'U) = t'(A + bU)$ and $\text{Var}(t'V|t'U) = t'Bt$ and $t'Bt$ is independent of U since B is independent of U . Hence $t'X_j$ for which $a_j \neq b b_j$ has a univariate normal distribution (Lukacs & Laha, 1964, p.123). But t is an arbitrary column vector of real constants. Therefore X_j has a multivariate normal distribution (Lukacs & Laha, 1964, p.30). This completes the proof.

Theorem 5

Let $X_1, \dots, X_n$ be n independently distributed symmetric stochastic matrices of order k each with each row having finite covariance matrices. Let $U = a_1 X_1 + \dots + a_n X_n$ and $V = b_1 X_1 + \dots + b_n X_n$ where a 's and b 's are scalars and $a_j b_j \neq 0$ for $j = 1, 2, \dots, n$. If $E(V|U) = A + bU$ and $\text{Cov}(V|U) = B$ where A is a matrix of constants, b is a scalar and B is positive definite and independent of U then X_j for which $a_j \neq b b_j$ has a $k(k+1)/2$ variate normal distribution.

Proof

Let T be an arbitrary column vector of constants. Consider $UT = a_1 X_1 T + \dots + a_n X_n T$ and $VT = b_1 X_1 T + \dots + b_n X_n T$. By theorem 4, $X_j T$ for which $a_j \neq b b_j$ has a multivariate normal distribution. This implies that each element in the vector $X_j T$ has a univariate normal distribution. Since T is arbitrary, each row vector in X_j has a k -variate normal distribution. But X_j is symmetric and hence X_j has a $k(k+1)/2$ variate normal distribution.

Theorem 6

Let $X_1, \dots, X_n$ be independently distributed column vectors of order k each, with finite covariance matrices. Let $U = A_1 X_1 + \dots + A_n X_n$ and $V = B_1 X_1 + \dots + B_n X_n$ where A 's and B 's are nonsingular matrices. If $E(V|U) = C + dU$ and $\text{cov}(V|U) = D$ where C is a vector, d is a scalar and D is a matrix independent of U , then X_j for which $A_j \neq d B_j$ has a multivariate normal distribution.

Proof

Consider $T'U$ and $T'V$ where T is an arbitrary constant column vector. By a similar argument as in theorem 4 we get that $T'AX_j$ has a univariate normal distribution. Since T is arbitrary AX_j has a multivariate normal distribution. But A is nonsingular and hence X_j has a multivariate normal distribution.

Theorem 7

Let $X_1, \dots, X_n$ be n independently distributed symmetric stochastic matrices of order k , with each row vector having finite covariance matrices. Let $\underline{U} = A_1 X_1 + \dots + A_n X_n$ and $\underline{V} = B_1 X_1 + \dots + B_n X_n$ where A 's and B 's are nonsingular matrices. If $E(\underline{V}|\underline{U}) = G + h \underline{U}$ and $\text{Cov}(\underline{V}|\underline{U}) = H$ where G is a matrix, h is a scalar and H is independent of $\underline{U}$, then X_j for which $A_j \neq h B_j$ has a $k(k+1)/2$ variate normal distribution.

The proof follows from theorems 5 and 6.

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REFERENCES

- LUKACS, E. (1962). On the characterization of a family of populations which includes the Poisson population. *Ann. Univ. Sci. Budapest. Sect. Mat.* 3-4, 159-175.
- LUKACS, E. & LAHA, R.G. (1964). *Applications of Characteristic Functions*. Griffin's Statistical Monographs, London.
- NEYMAN, J. (1951). Existence of consistent estimates of the directional parameter in a linear structural relation between two variables. *Ann. Math. Statist.* 22, 497-512.
- STEYN, H.S. (1957). On regression properties of discrete systems of probability functions. *Indag. Math.* 19, 119-127.
- STEYN, H.S. (1960). On regression properties of multivariate probability functions of Pearson types. *Indag. Math.* 22, 303-311.

WOLFOWITZ, J. (1952). Consistent estimators of the parameters of a linear structural relation. *Skand. Aktuarietidskrift*. 35, 132-151.

ZINGER, A.A. & LINNIK, Yu (1957). On a class of differential equation with an application to certain questions of regression theory. *Vestnik Leningrad. Univ.* 12, 121-130.

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